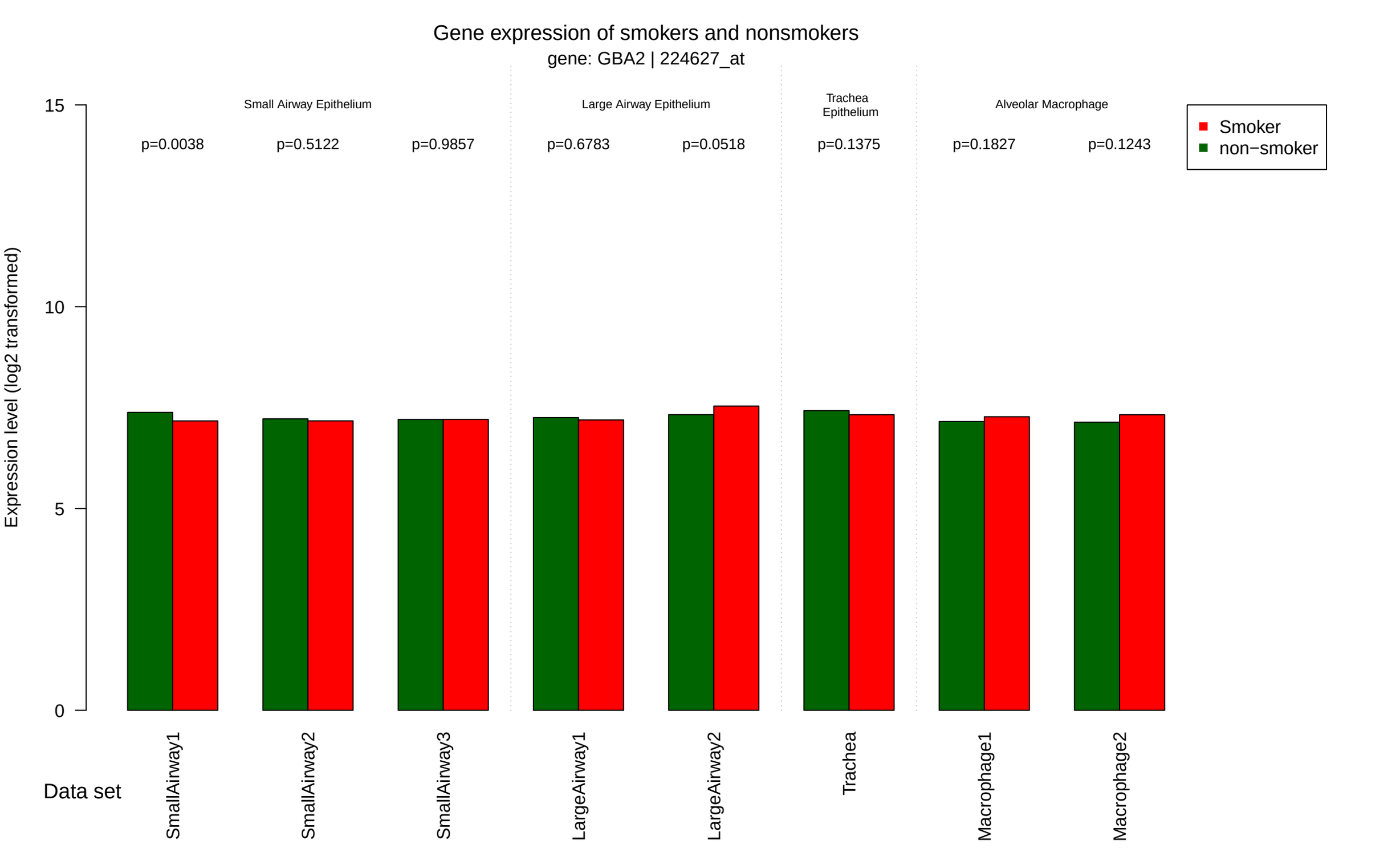
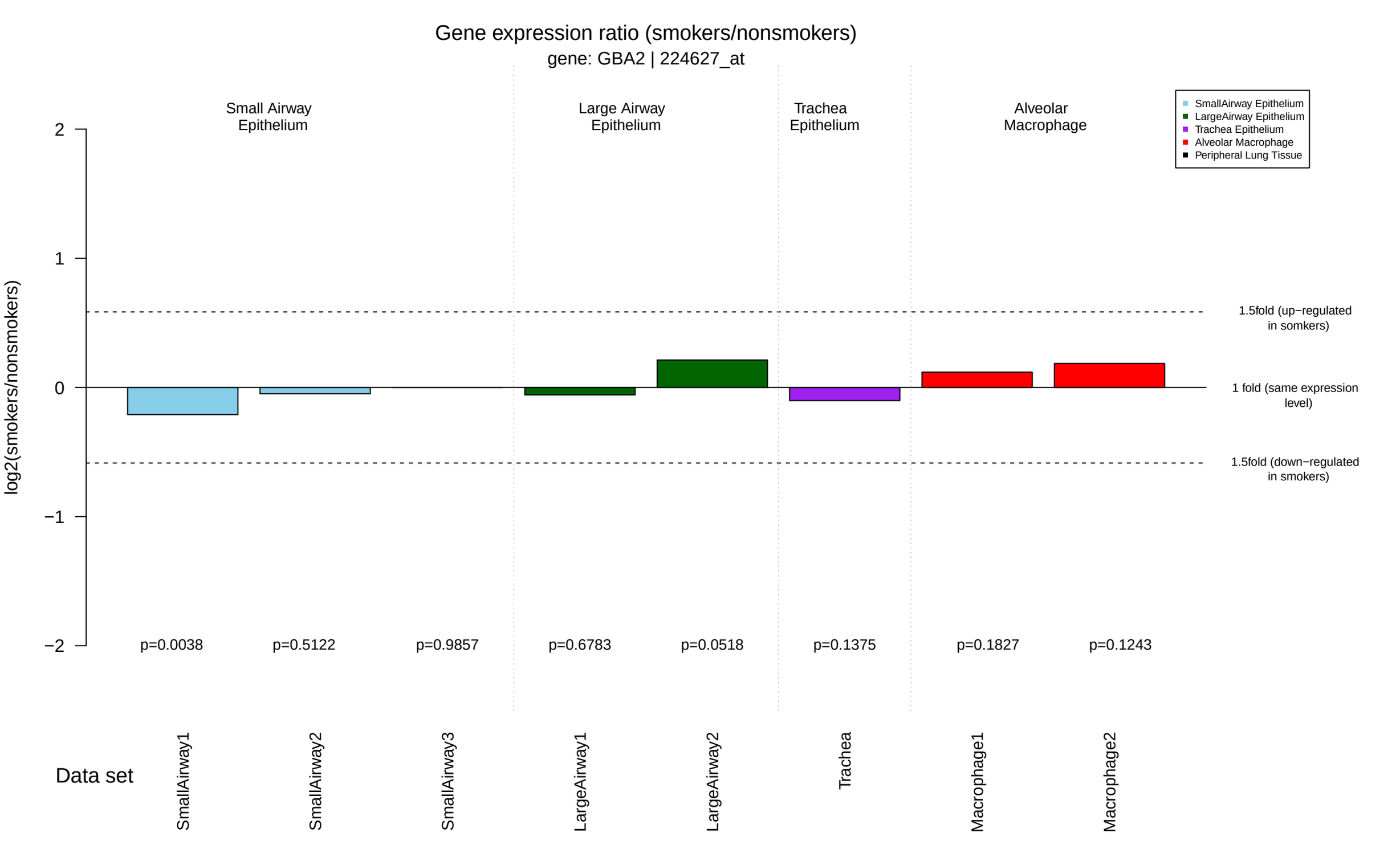
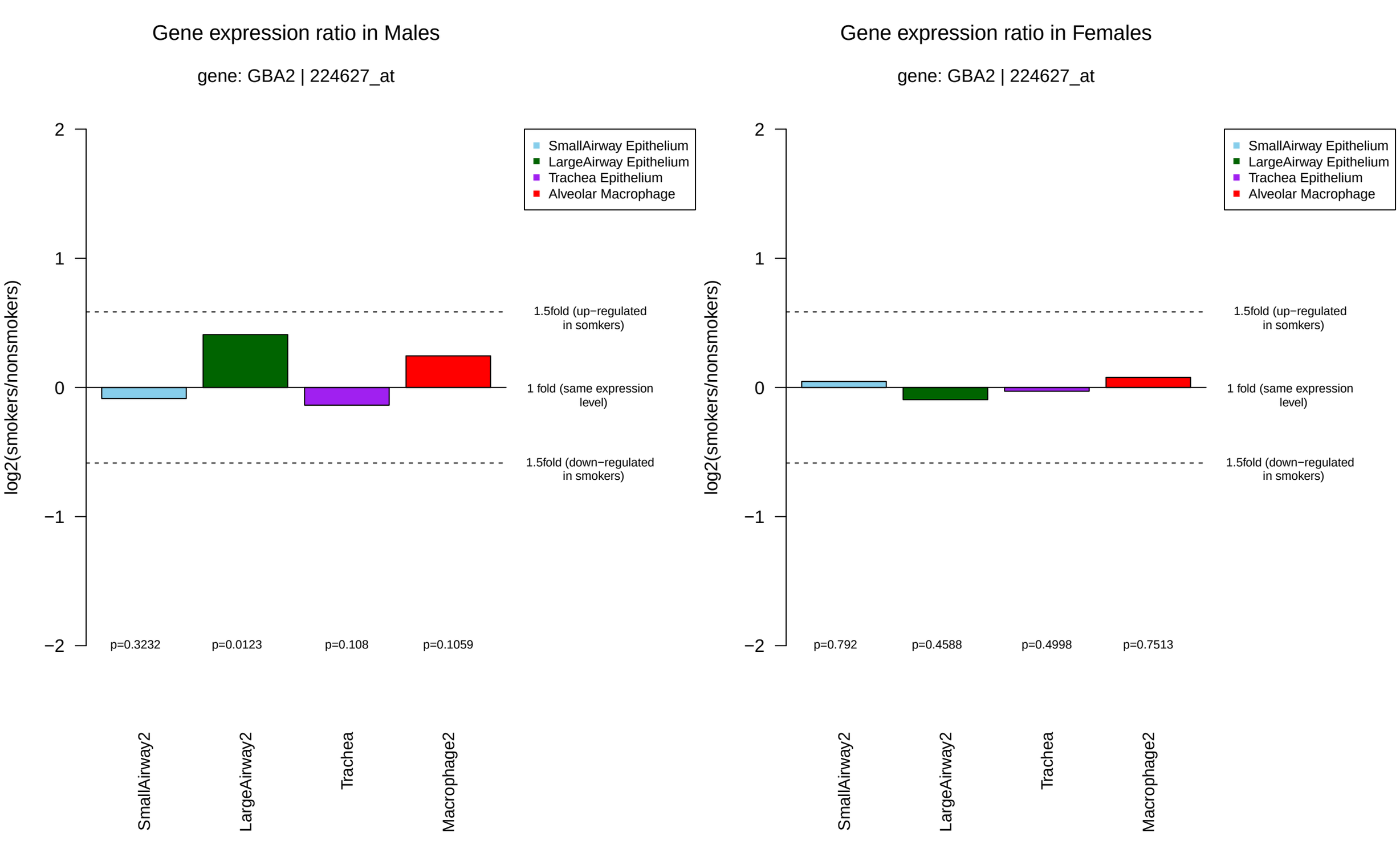
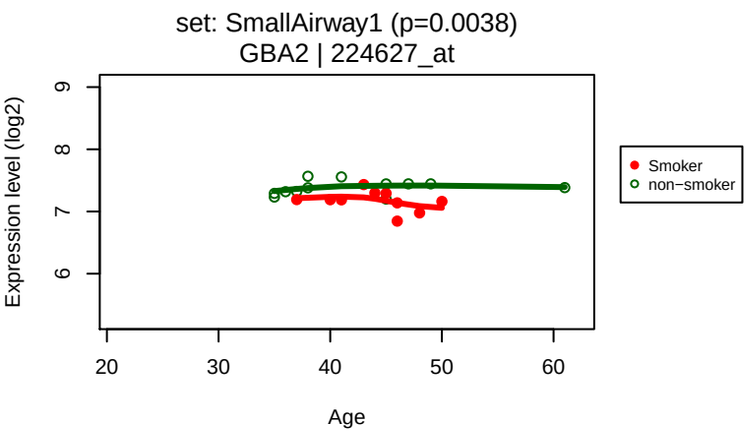
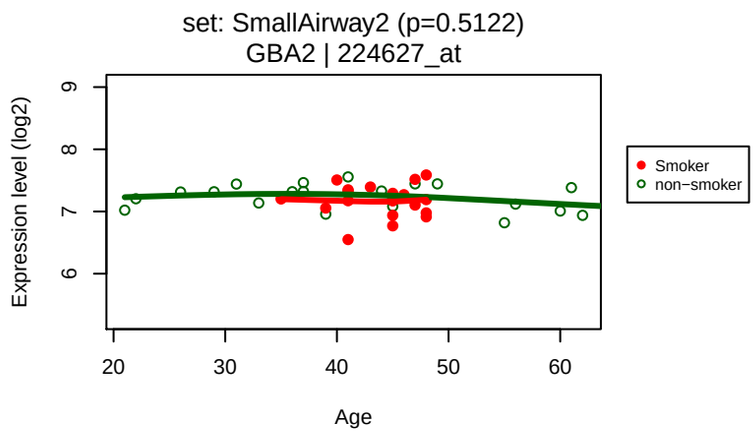
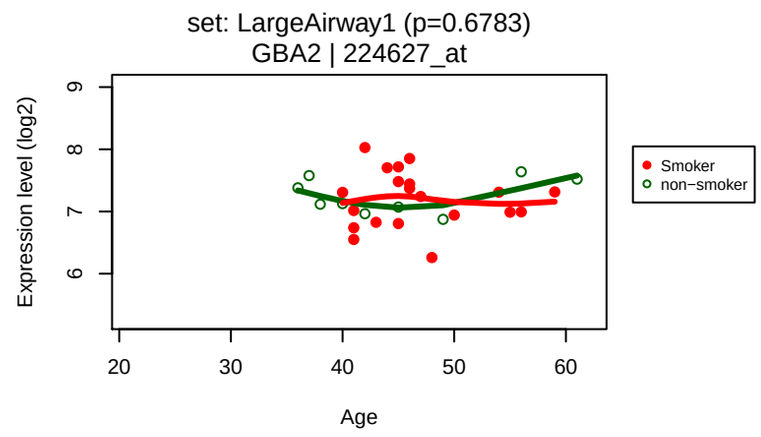

Gene expression of smokers and nonsmokers
gene: GBA2 | 224627_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0038
p=0.5122
p=0.9857
p=0.6783
p=0.0518
p=0.1375
p=0.1827
p=0.1243
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

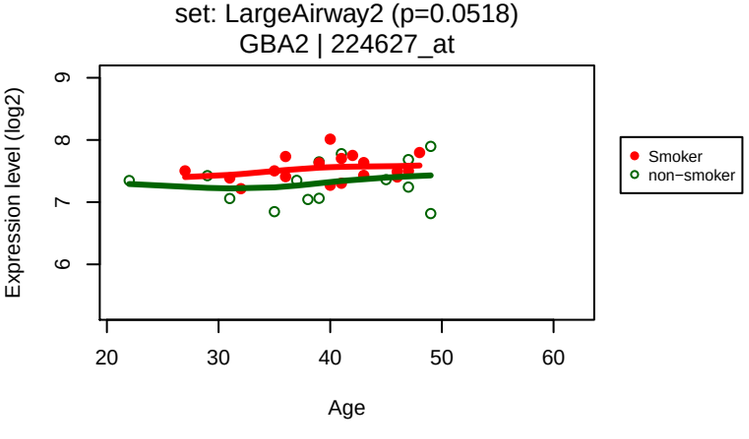

Gene expression ratio (smokers/nonsmokers)
gene: GBA2 | 224627_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0038
p=0.5122
p=0.9857
p=0.6783
p=0.0518
p=0.1375
p=0.1827
p=0.1243
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

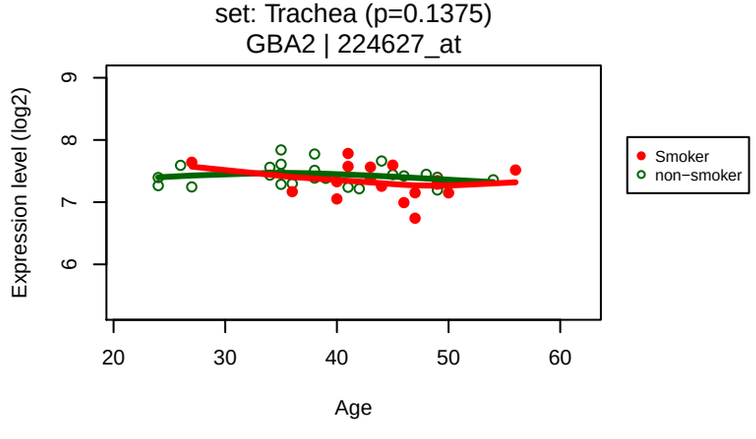

Gene expression ratio in Males
gene: GBA2 | 224627_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3232
p=0.0123
p=0.108
p=0.1059
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GBA2 | 224627_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.792
p=0.4588
p=0.4998
p=0.7513
SmallAirway2
LargeAirway2
Trachea
Macrophage2

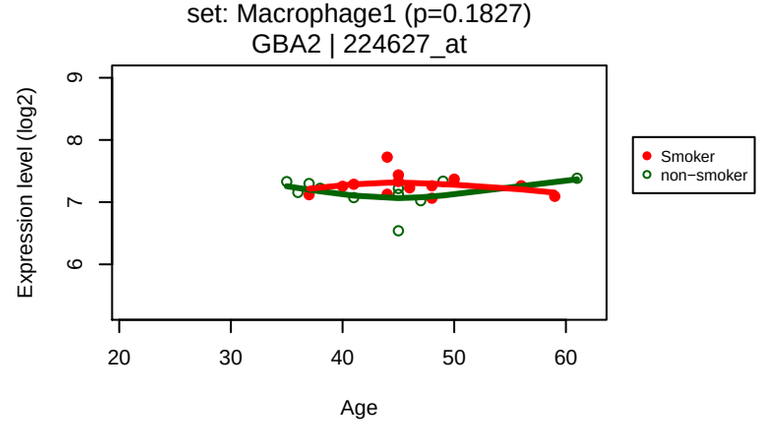

set: SmallAirway1 (p=0.0038)
GBA2 | 224627_at
Expression level (log2)
Age
Smoker
non−smoker

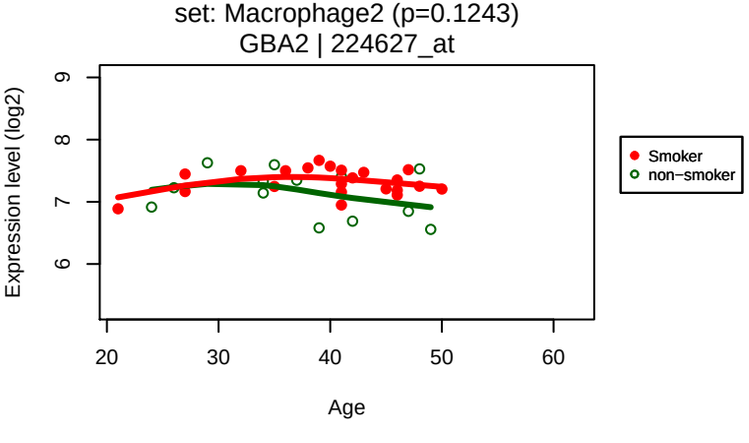

set: SmallAirway2 (p=0.5122)
GBA2 | 224627_at
Expression level (log2)
Age
Smoker
non−smoker

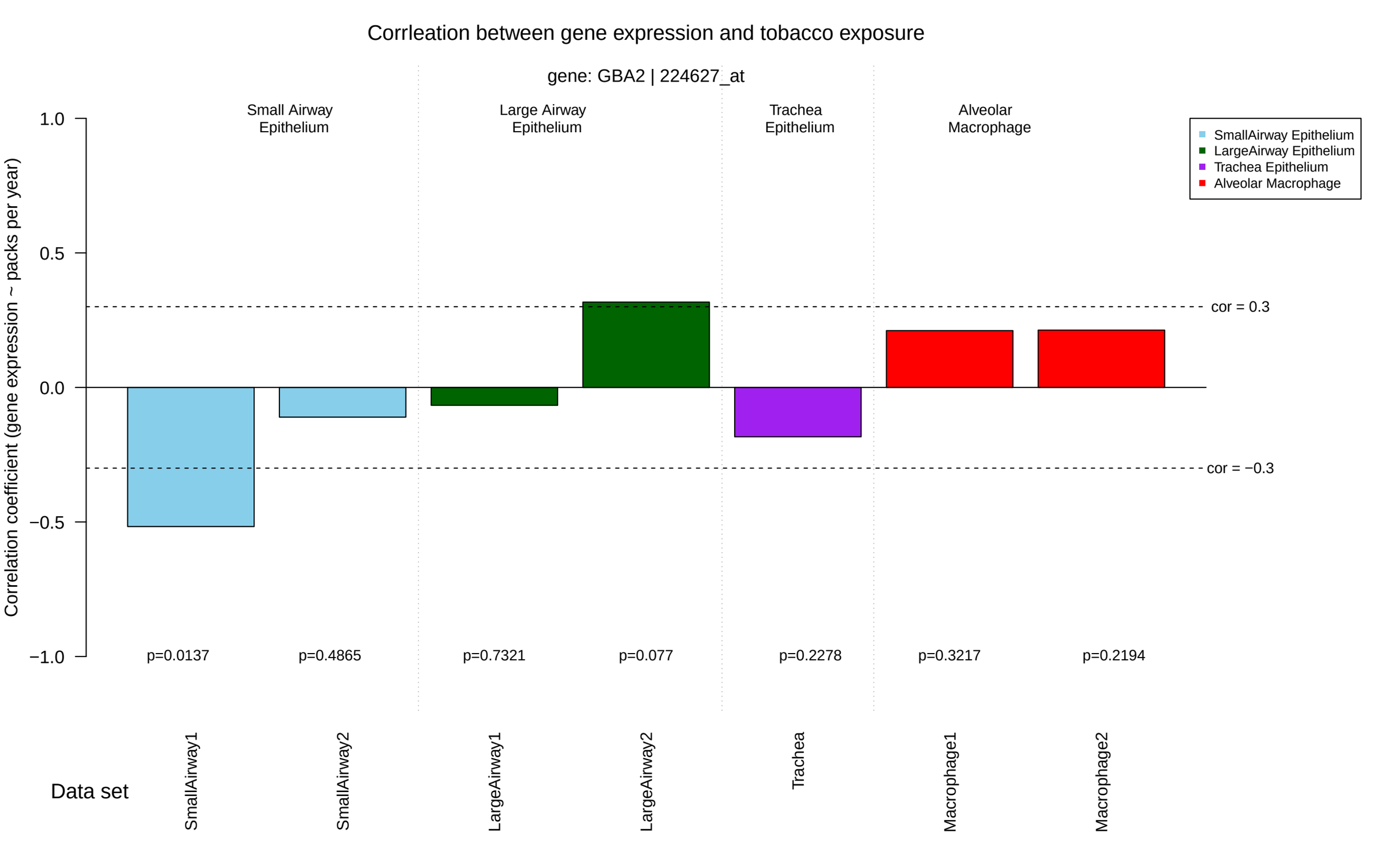

set: LargeAirway1 (p=0.6783)
GBA2 | 224627_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0518)
GBA2 | 224627_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1375)
GBA2 | 224627_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1827)
GBA2 | 224627_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1243)
GBA2 | 224627_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GBA2 | 224627_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0137
p=0.4865
p=0.7321
p=0.077
p=0.2278
p=0.3217
p=0.2194
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2